Project - Drought Matrices

Drought Detection Based on Varying Treshlolds on SPI & SPEI

Thresholds to define the beginning and end of a drought event

We use a potential definition of drought events based on SPI and SPEI to determine drought events. Here we use ERA5 data from 2001 to 2023 to determine how Argentina would face drought events based on GWSC's potential definition of drought. Our goal here is to determine how these definitions will pick up drought events under a given climate scenario, and later evaluate them against true drought events to measure how well our definition performs.

1 Thresholds Definition 01

Cumulative: drought begins when drought index is < 0 for at least 2 months and ends when the sum of the negative values = the sum of the positive values. This method allows for a drought event to continue even with months of above normal rainfall. In practice, drought impacts don't end with the first rain and it may take many months of above normal rain to recover from a drought event.

Based on above definition in Argentina's Drought event from 2001-2023 are summarized as follows:

Drought Event Summary : Definition 1

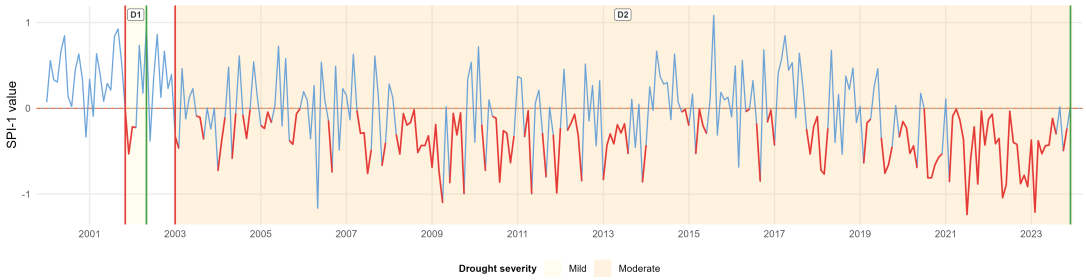
SPI time frame	Mean Duration (Months)	Mean Peak SPI (-)	Mean cumi.sum SPI (+)	Net SPI during event	Mean cumi.sum SPI (-)	Total drought event
SPI1	129.5	-0.8844528	13.572436	-20.69742	-21.65214	2
SPI3	128.5	-0.9403690	10.106607	-31.80426	-32.56423	2
SPI6	248.0	-1.5701083	13.673855	-81.48865	-81.48865	1
SPI12	243.0	-1.7794856	9.740094	-100.99038	-100.99038	1

Multi-scale SPI Drought Detection

Cumulative run theory | Start: SPI < 0 | End: cumulative SPI ≥ 0 | Min consecutive months: 2

SPI-1 — Drought Events | 2 detected

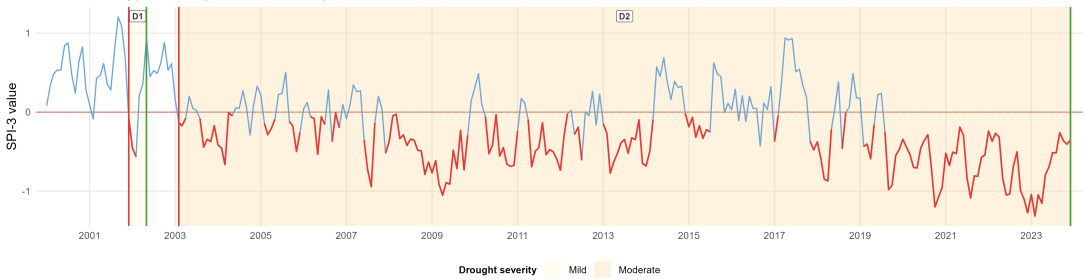
Cumulative run theory | Start: SPI < 0 | End: cumulative SPI ≥ 0 | Min consecutive months: 2



Shading = severity class (Mild / Moderate / Severe / Extreme)

SPI-3 — Drought Events | 2 detected

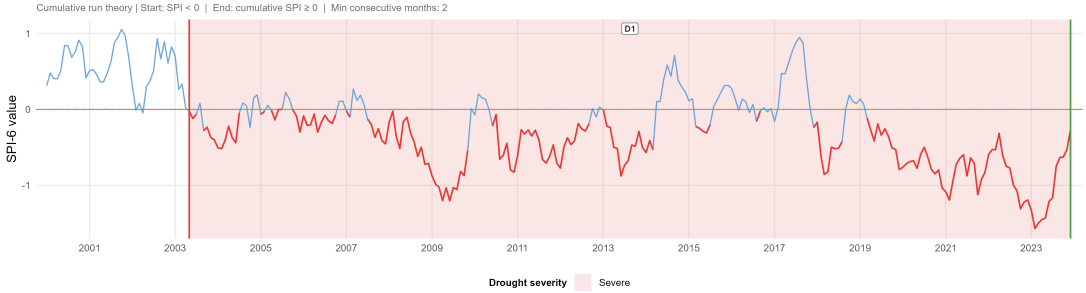
Cumulative run theory | Start: SPI < 0 | End: cumulative SPI ≥ 0 | Min consecutive months: 2



Shading = severity class (Mild / Moderate / Severe / Extreme)

SPI-6 — Drought Events | 1 detected

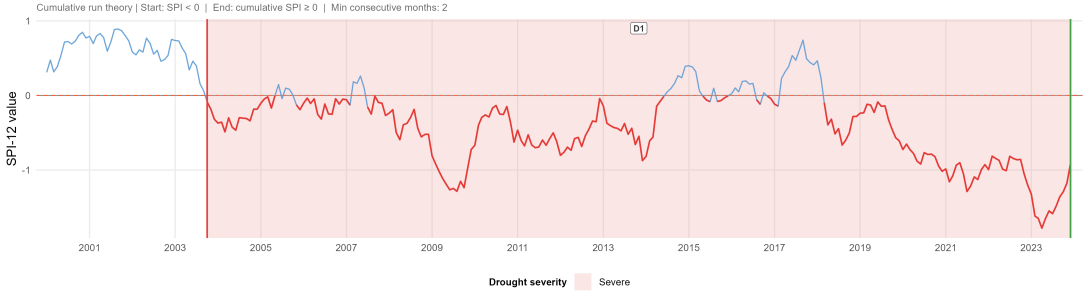
Cumulative run theory | Start: SPI < 0 | End: cumulative SPI ≥ 0 | Min consecutive months: 2



Shading = severity class (Mild / Moderate / Severe / Extreme)

SPI-12 — Drought Events | 1 detected

Cumulative run theory | Start: SPI < 0 | End: cumulative SPI ≥ 0 | Min consecutive months: 2



Shading = severity class (Mild / Moderate / Severe / Extreme)

2 Thresholds Definition 02

–0.5: Number of consecutive months in which the drought index remained below –0.5. This method is a traditional way to define drought with a commonly accepted threshold. To initiate a drought event, there should be a minimum of two consecutive months where the SPI remains below –0.5, and once it returns to an SPI value greater than –0.5, the drought event ends. It does not need to return to a positive value — an SPI value that is still negative but higher than –0.5 would call off the drought event.

Drought Event Summary : Definition 2

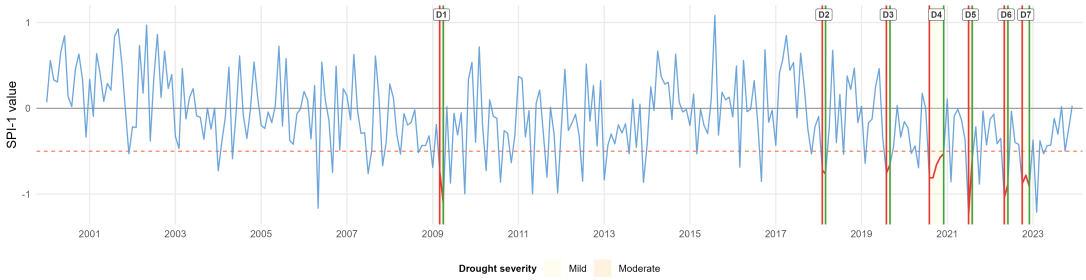
SPI time frame	Mean Duration (Months)	Mean Peak SPI (-)	Mean cumi.sum SPI (+)	Net SPI during event	Mean cumi.sum SPI (-)	Total drought event
SPI1	2.571429	-0.9471190	0	-2.063664	-2.063664	7
SPI3	5.083333	-0.9204976	0	-3.997188	-3.997188	12
SPI6	7.166667	-0.8561563	0	-5.876017	-5.876017	12
SPI12	18.800000	-1.0816594	0	-17.102820	-17.102820	5

Multi-scale SPI Drought Detection

Simple threshold | Start: SPI < -0.5 | End: SPI ≥ -0.5 | Min consecutive months: 2

SPI-1 — Drought Events | 7 detected

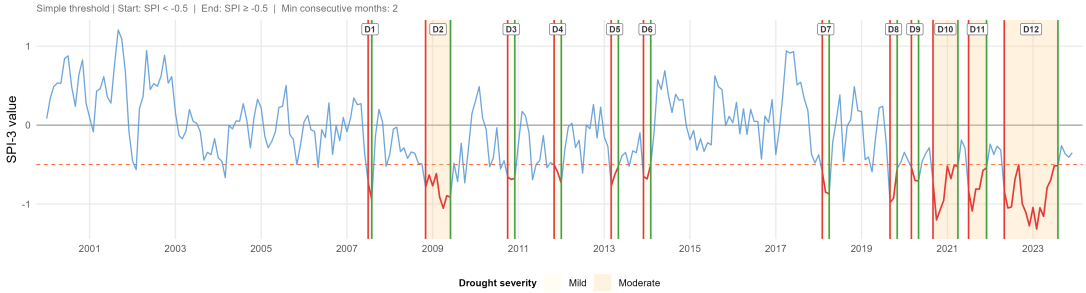
Simple threshold | Start: SPI < -0.5 | End: SPI ≥ -0.5 | Min consecutive months: 2



Shading = severity class (Mild / Moderate / Severe / Extreme)

SPI-3 — Drought Events | 12 detected

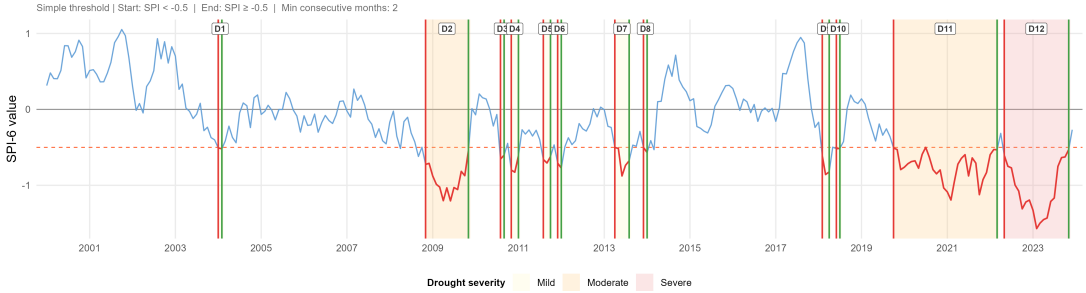
Simple threshold | Start: SPI < -0.5 | End: SPI ≥ -0.5 | Min consecutive months: 2



Shading = severity class (Mild / Moderate / Severe / Extreme)

SPI-6 — Drought Events | 12 detected

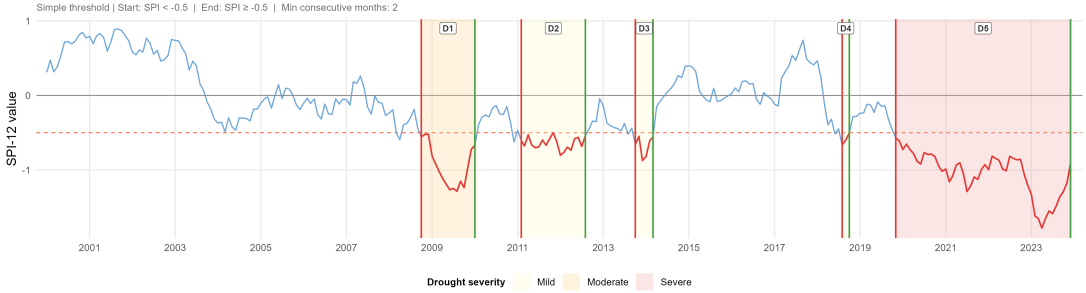
Simple threshold | Start: SPI < -0.5 | End: SPI ≥ -0.5 | Min consecutive months: 2



Shading = severity class (Mild / Moderate / Severe / Extreme)

SPI-12 — Drought Events | 5 detected

Simple threshold | Start: SPI < -0.5 | End: SPI ≥ -0.5 | Min consecutive months: 2



Shading = severity class (Mild / Moderate / Severe / Extreme)

3 Thresholds Definition 03

–1: Number of consecutive months in which the drought index remained below –1. This method is a traditional way to define drought with a commonly accepted threshold. To initiate a drought event, there should be a minimum of two consecutive months where the SPI remains below –1, and once it returns to an SPI value greater than –1, the drought event ends. It does not need to return to a positive value — an SPI value that is still negative but higher than –1 would call off the drought event.

Drought Event Summary : Definition 3

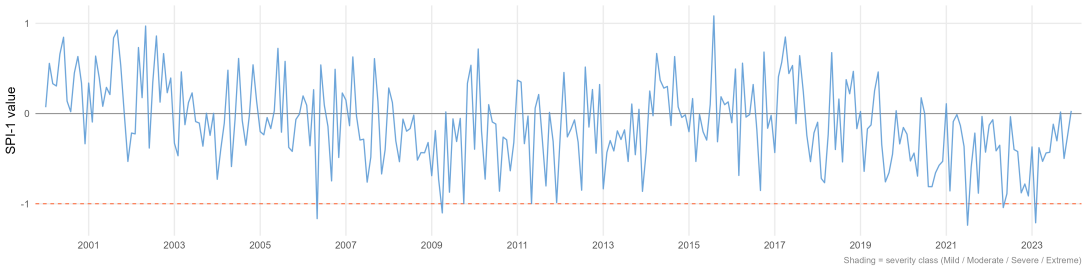
SPI time frame	Mean Duration (Months)	Mean Peak SPI (-)	Mean cumi.sum SPI (+)	Net SPI during event	Mean cumi.sum SPI (-)	Total drought event
SPI1	0.000000	0.000000	0	0.000000	0.000000	0
SPI3	3.333333	-1.189308	0	-3.768431	-3.768431	3
SPI6	6.666667	-1.323130	0	-8.108440	-8.108440	3
SPI12	7.000000	-1.378096	0	-9.060880	-9.060880	4

Multi-scale SPI Drought Detection

Simple threshold | Start: SPI < -1 | End: SPI ≥ -1 | Min consecutive months: 2

SPI-1 — Drought Events | 0 detected

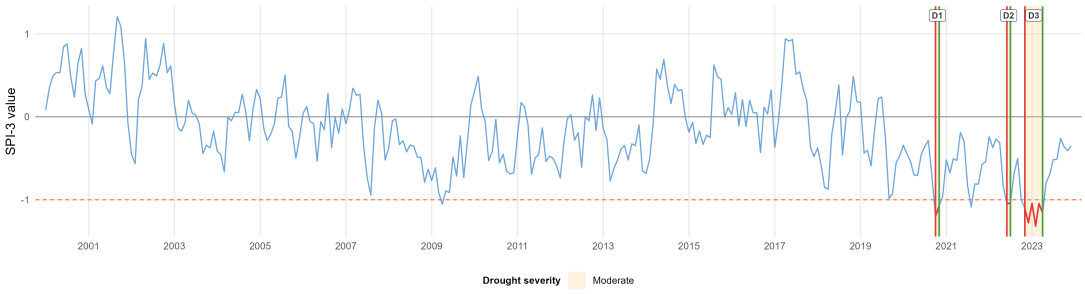
Simple threshold | Start: SPI < -1 | End: SPI ≥ -1 | Min consecutive months: 2



Shading = severity class (Mild / Moderate / Severe / Extreme)

SPI-3 — Drought Events | 3 detected

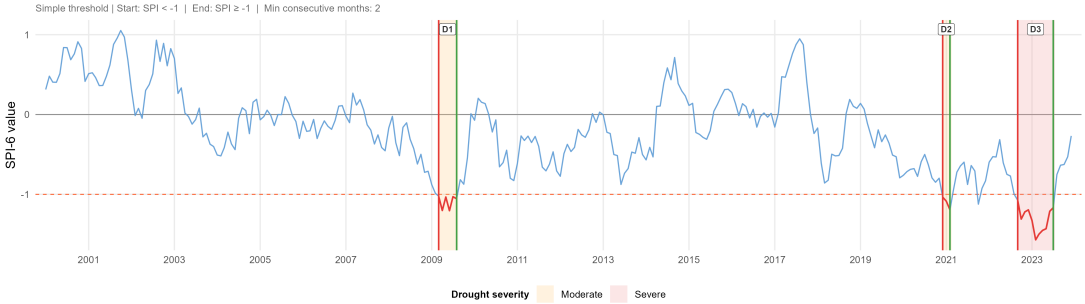
Simple threshold | Start: SPI < -1 | End: SPI ≥ -1 | Min consecutive months: 2



Shading = severity class (Mild / Moderate / Severe / Extreme)

SPI-6 — Drought Events | 3 detected

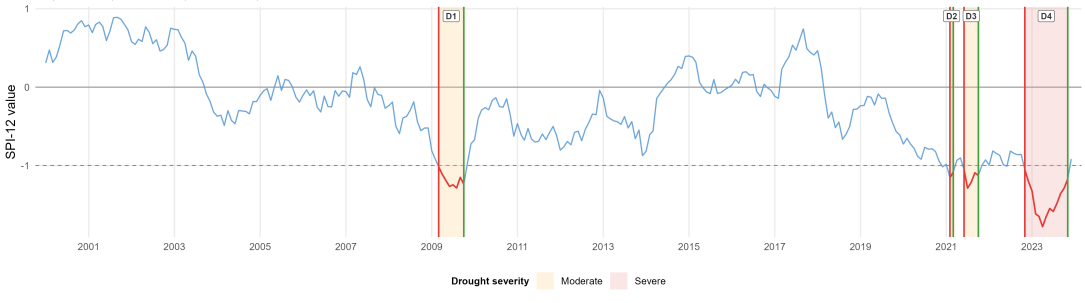
Simple threshold | Start: SPI < -1 | End: SPI ≥ -1 | Min consecutive months: 2



Shading = severity class (Mild / Moderate / Severe / Extreme)

SPI-12 — Drought Events | 4 detected

Simple threshold | Start: SPI < -1 | End: SPI ≥ -1 | Min consecutive months: 2



Shading = severity class (Mild / Moderate / Severe / Extreme)

Based on SPEI

4 Thresholds Definition 01

Drought Event Summary : Definition 1

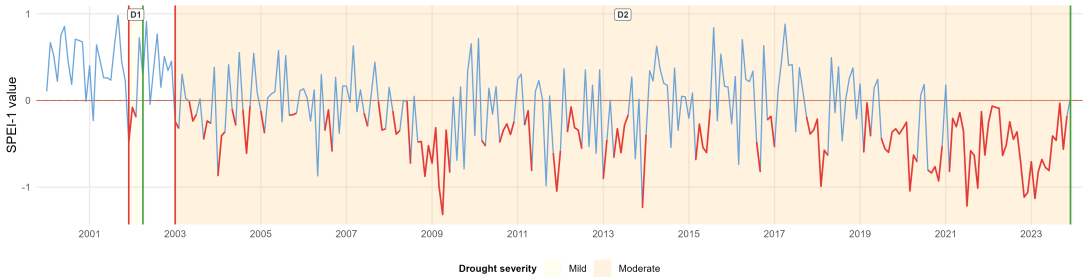
SPEI time frame	Mean Duration (Months)	Mean Peak SPEI (-)	Mean cumi.sum SPEI (+)	Severity (+) - (-)	Mean cumi.sum SPEI (-)	Total drought event
SPEI1	128.5	-0.8998495	11.985383	-26.33168	-26.84935	2
SPEI3	127.0	-0.9360101	7.784121	-39.16382	-39.53554	2
SPEI6	247.0	-1.5693762	10.185118	-98.21153	-98.21153	1
SPEI12	243.0	-1.6174273	4.497822	-119.39140	-119.39140	1

Multi-scale SPEI Drought Detection

Cumulative run theory | Start: SPEI < 0 | End: cumulative SPEI ≥ 0 | Min consecutive months: 2

SPEI-1 — Drought Events | 2 detected

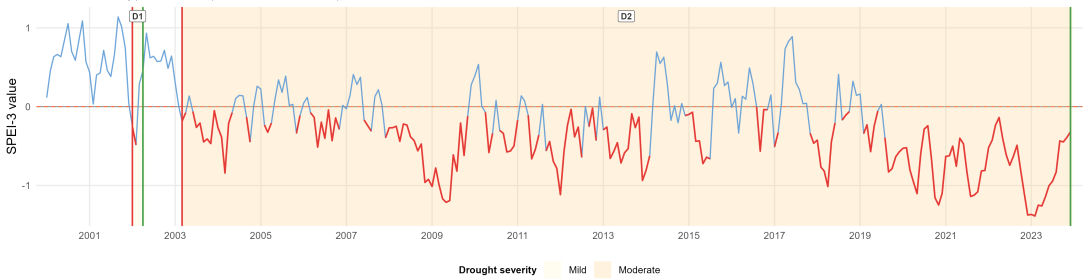
Cumulative run theory | Start: SPEI < 0 | End: cumulative SPEI ≥ 0 | Min consecutive months: 2



Shading = severity class (Mild / Moderate / Severe / Extreme)

SPEI-3 — Drought Events | 2 detected

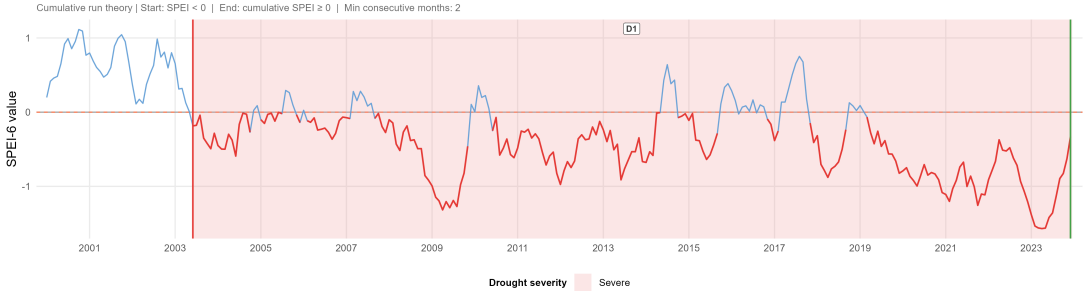
Cumulative run theory | Start: SPEI < 0 | End: cumulative SPEI ≥ 0 | Min consecutive months: 2



Shading = severity class (Mild / Moderate / Severe / Extreme)

SPEI-6 — Drought Events | 1 detected

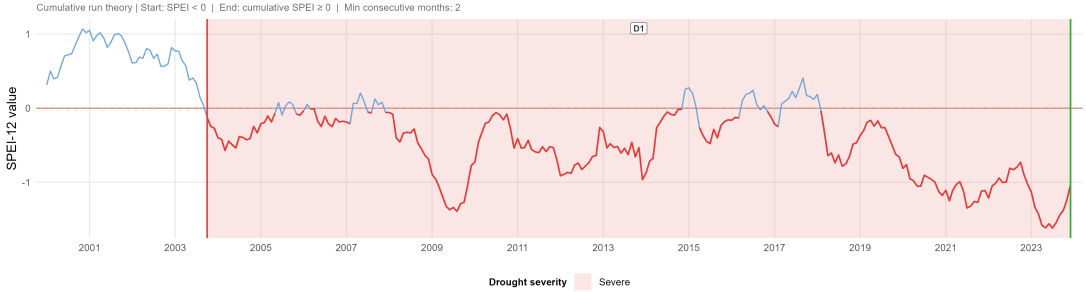
Cumulative run theory | Start: SPEI < 0 | End: cumulative SPEI ≥ 0 | Min consecutive months: 2



Shading = severity class (Mild / Moderate / Severe / Extreme)

SPEI-12 — Drought Events | 1 detected

Cumulative run theory | Start: SPEI < 0 | End: cumulative SPEI ≥ 0 | Min consecutive months: 2



Shading = severity class (Mild / Moderate / Severe / Extreme)

Drought Event Summary : Definition 2

SPI time frame	Mean Duration (Months)	Mean Peak SPEI (-)	Mean cumi.sum SPEI (+)	Severity (+) - (-)	Mean cumi.sum SPEI (-)	Total drought event
SPEI1	3.454545	-0.9438949	0	-2.708204	-2.708204	11
SPEI3	5.071429	-0.9452013	0	-4.283885	-4.283885	14
SPEI6	9.200000	-0.9363009	0	-8.085959	-8.085959	10
SPEI12	15.142857	-0.9759525	0	-14.081505	-14.081505	7

5 Thresholds Definition 02



Drought Event Summary : Definition 3

SPEI time frame	Mean Duration (Months)	Mean Peak SPEI (-)	Mean cumi.sum SPEI (+)	Severity (+) - (-)	Mean cumi.sum SPEI (-)	Total drought event
SPEI1	2.00	-1.114862	0	-2.176112	-2.176112	1
SPEI3	4.25	-1.247059	0	-5.074087	-5.074087	4
SPEI6	6.25	-1.336523	0	-7.831286	-7.831286	4
SPEI12	8.00	-1.331140	0	-9.976927	-9.976927	5

6 Thresholds Definition 03

